

Laxy Studio Function List

Laxy Studio — Function List

Hierarchy: Module → Sub-Module → Function → Feature → Ticket → Remark

Scope: Laxy Studio (B2B creator platform). Player PWA = separate table.

References: Local docs 01–06 + Confluence: System Architecture, Development Roadmap

Proposed Complete Table

Module	Sub-Module	Function	Feature	Phase	Ticket	Remark
Infrastructure	Auth & RBAC	SSO / OAuth Login	- Google / Apple social login - Firebase Authentication	Phase 1		ok, just like what TMX did, so we also need use resend.com (need another account for free plan)
		JWT / Session Management	- Issue & refresh access tokens - Token expiry policy	Phase 1		will implement
		Role-Based Access Control	- Roles: Laxy Super Admin, Client Admin, Client Editor - Permission matrix per role	Phase 1		yes, can implement by Firebase Auth, but need you define the permission matrix

					HL:  System Architecture 4. User Roles & Permissions s Use FireCMS to implement
	Cloud Storage	Asset Storage	- Support PDF, image, video, audio Tenant scoped isolation	Phase 1	GCP Cloud Storage Tenant scoped, so it should be equivalent to “Client” in RBAC. You decide align the term or not HL: Client should also control asset ⇒ Tenant > Client Admin
		Output Storage	- Store generated audio, SRT, slideshow assets - Versioned output	Phase 2	How the versioning pattern be? just v1,2,3? HL: please design

					Need more detail > Phase 2
	Database	Relational DB (PostgreSQL)	Users, Tenants, Permissions, Billing, Prompts	Phase 1	PostgreSQL, yes Again, Users, Client Admin terms are bit confusing? HL: let's discuss
		Document DB (Firestore)	Content, Spots, Scripts, AudioFiles (real-time sync)	Phase 1	Just Firestore.
		Vector DB	AI embeddings for Chatbot & Q&A semantic search	Phase 3	Should not be in Phase 1 Pinecone / Weaviate — Phase 3 dependency HL: yes phase 3
	API Gateway	API Routing & Auth Middleware	Control by ADK pipeline	Phase 1	Most control by ADK pipeline on Phase 1, can set gateway

					in next phases All AI ops return taskId for async polling @Sun draw a clear relationships between API
			• Single entry point, Firebase ID token validation - Base URL: api.laxy.travel/v1	Phase 2	
		Rate Limiting & Versioning	• Per-tenant rate limits - Versioned API contracts	Phase 1	confirm set in Phase 1? it is a question, since for Gemini and 3rd party side, also has rate limit. so need some testing first

					Per-tenant rate limits, the first assumption is we can have much huge rate limit in Gemini side first Display token spending for internal evaluation first
	AI Orchestrator	Event Queue (Pub/Sub)	- Async routing between Studio and Orchestrator - Key events: ASSET_UPLOADED → CONTENT_PUBLISHED	Phase 1	ADK with Event Queue (Pub/Sub)
		Prompt Router	- Route prompts by function via Prompt Library - Vendor-agnostic model adapter	Phase 1	I think the prompt node in ADK is static in Phase 1 which set to hard coded to specific LLM Decoupled — swap AI

						providers without code change HL: Hard coded to Gemini 3.1 pro preview? Or could define on which model by on each prompt in library?
		NLP / OCR Service	Text/PDF/Excel/Word only	Phase 1		yes, and faced rate limit before, need apply for larger limit HL: Cost eatimation. Have to implement in different phase?
			Images	Phase 2		
		Script & Translation Service	Gemini Pro: generate and translate scripts	Phase 1		yes
		TTS Service	Gemini 2.5 Pro TTS: generate speech audio	Phase 1		yes
		Image Generation Service	Gemini / Imagen: avatars,	Phase 2		yes HL: Do in phase 2

			slideshows, stamp art			
	CI/CD & Environments	Deployment Pipeline	- GitHub Actions: Studio, Player, API, Orchestrator pipelines - Environments: Dev / Staging / Production	Phase 1		laxy-pipelines (Orchestrator pipelines) laxy-studio (included FireCMS(laxy - studio/admin)) laxy-guide (React Frontend Player) so please confirm the URL: [env]-studio.laxy.travel [env]-guide.laxy.travel/{guideID} confirm the flow: DEV, Staging, PROD

					data are totally isolated No data migration at normal case. Expect tenant only do operation in PROD studio? Different env TOTALLY separate
		Monitoring & Logging	- Cloud Logging, Monitoring, Error Reporting - Audit logs for all roles actions	Phase 1	Phase 1, by default show what ADK default provide? Internal Laxy team HL: Audit logs for all roles actions → All users
	Security	Data Protection	- Encryption at rest and in transit (TLS) - Signed URLs for file access	Phase 2	Prompts fetched server-side only, yes, prompts

			• Prompts fetched server-side only • GCP Secret Manager for secrets		only exists in ADK , not in studio No handle signed URLs in Phase 1 No special GCP Secret Manager, just follow Firestore practice in Phase 1
			• Signed URLs for file access • GCP Secret Manager for secrets	Phase 2	
Super Admin	Prompt Library	Prompt Management	• Create, version, activate AI prompts per function • Prompt tagging by module • Only Client Super Admin can edit selected prompts	Phase 1	yes, Decouple AI tuning from code deployment HL: Only Super Admin could edit prompt
		A/B Prompt Testing	• Run prompt variants, compare output quality	Phase 2	Option 1: Same Pipeline + Different Variable/Override Values

					Option 2: Different Pipelines (Recommended for Bigger Variants) Multi prompt under same module for selection
	User & Tenant Management	Tenant CRUD	• Create, read, update, deactivate tenant accounts	Phase 1	 Recommen ded high-lev el architectur e for Laxy St udio Phase 1 MVP laxy-studio (React frontend + FireCMS) - Main app: studio.laxy.travel - Admin: /admin route with FireCMS - Wizard for Guide creation
		User CRUD	• Manage users within tenants • Assign roles (Client Admin / Client Editor)	Phase 1	Handle in FireCMS

		Subscription Tier Override	Manually adjust tier/quota for a tenant	Phase 1.5		Handle in FireCMS
		Module Dashboard	- Enable/disable modules per tenant - Controls Studio and Player visibility	Phase 2		Handle in FireCMS
	Platform Config & Analytics	Feature Flags	Enable/disable features per tenant or globally	Phase 1		Handle in FireCMS
		Platform Analytics	Usage metrics and revenue overview across all tenants	Phase 3		Handle in FireCMS
Laxy Studio	Remark		AI would make mistake. Please verify before publish			
	Member Management	Login	- Social login (Google, Apple) - Email & password login - Email OTP verification	Phase 1		Handle in Studio App
		Registration	- Tenant sign-up form - Email verification - Onboarding walkthrough	Phase 1		Handle in Studio App

		Forget Password	- Reset via email link - OTP expiry policy	Phase 1		Handle in Studio App
		Account Management	- Change email, password, display name - Profile avatar upload	Phase 1		Handle in FireCMS
		Tenant Information	- Company name, logo, billing info - Timezone & default language - Default audio settings (Character, Voice, Context) used by Quick Process	Phase 1		Handle in FireCMS
		User Management	- Invite users by email - Assign / change roles - Deactivate / remove users	Phase 2		Handle in FireCMS
	Asset Management	Asset Upload	- Drag & drop upload (PDF, image) - Bulk upload with progress indicator - URL / raw text input as content source	Phase 1		Handle in Studio App (Wizard) Max file size TBD

					HL: 100M
			• Drag (video, audio)	Phase 2	
		Asset Library	• Browse, search, preview assets • Delete replace assets • Storage usage indicator	Phase 1	Handle in Studio App (Wizard) HL: no relpace
		Asset Tagging	• Tag assets for retrieval across modules		Not in Phase 1? HL: no tag need, I thnk
	Basic Information	Entity Configuration	• Venue name, address, GPS coordinates • Map image, cover image • Operating hours, website, phone • Supported languages • Core language • Enabled modules • Item field config (e.g. title, year...)	Phase 1	Handle in Studio App (Wizard) Top-level setup before any module
		Layout Style Selection	• Select from pre-designed Player UI templates (Phase 1)	Phase 1	Handle in Studio App (Wizard)

			• Live preview of selected layout			
		AI Layout Generation	• Generate custom UI from brand reference + prompt	Phase 3		Phase 2
		Function / Module Selection	• Guide	Phase 1		Managed via Module Dashboard
			• Q&A • Chatbot • Stamp Hunt	Phase 3		
	Guide — Step 1: Ingestion	Content Upload	• Select files already in Asset Data Lake, OR upload new files here (New uploads automatically saved to Asset Data Lake • Or provide basic entity info manually • Select core language	Phase 1		Handle in Studio App (Wizard) to Firebase Files always centralised in Asset Data Lake regardless of entry point

		AI Metadata Extraction	• AI extracts: Title, Artist, Period, Material, Dimensions, Highlight, Cultural Designation, and more • Add number to items (if any) • Drag-and-drop reorder	Phase 1		Handle in Studio App (Wizard) and ADK Gemini Multimodal OCR/NLP
		Human Gate 1 — Data Review	• Client reviews and corrects extracted metadata inline • Approve to proceed	Phase 1		Handle in Studio App (Wizard) and ADK
			• Export to Excel for batch review	Phase 2		
	Guide — Step 2: Deep Research <i>(Phase 2)</i>	Deep Research Activation	• Optional: only if Deep Research module is enabled • Degree of Research selection: Low / Medium / High (higher degree = broader crawl but higher risk of inaccurate content) • AI crawls: art critiques,	Phase 2		Phase 2 — Epic 7

			historical backgrounds, artist biographies, conservation notes - ⚠️ Client must be clearly informed: higher depth increases inaccuracy risk mandatory Human Gate 2 review before proceeding			
		Human Gate 2 — Research Review	• Client reviews AI-researched content for accuracy - Approve to proceed	Phase 2		Skipped in Quick Process
	Guide — Step 3: Script Generation	AI Script Generation	• AI generates polished rewritten description per spot • Based on approved metadata	Phase 1		Handle in Studio App (Wizard) and ADK Multiple variants = different end-user depth options
			• AI generates audience-variant scripts per spot:	Phase 2		

			◦ Kids, Academic, Quick (1-min), Professional, Brief ◦ In-Depth / Enthusiast (requires Deep Research, Phase 2) ◦ V1, V2, V3 variants per type			
		Image-Spot Mapping	• Assign images/videos from asset library to each spot —AI auto— suggests media —Manual— override	Phase 1		??? how the logic would be??? I guess it is hard to get good result HL: AI assign too complicated
		Human Gate 3 — Script Review	• Client reviews AI-generated scripts per spot - Approve individually or in bulk • Fast Track: bypass gate for trusted/simple items	Phase 1		Fast Track = spot-level skip; Quick Process = full pipeline bypass Handle in Studio App (Wizard) and ADK

	Guide — Step 4: Translation	AI Translation	• AI translates approved base script to all configured languages (EN, JP, KR, ZH-TW, ZH-CN, FR, etc.) • Culturally appropriate phrasing	Phase 1		Handle in Studio App (Wizard) and ADK Gemini Pro
		Human Gate 4 — Translation Review	• Client reviews translated scripts • Approve to proceed	Phase 1		Handle in Studio App (Wizard) and ADK Skipped in Quick Process
			• Export to Excel for external translators	Phase 2		
	Guide — Step 5: Audio	Character Selection	• Select from Character Library (voice persona: name, role, avatar, personality, speech patterns) • Client Admin: create new character via meta-prompt	Phase 1		Handle in Studio App (Wizard) and ADK Check this Youtube link before these magic happen  那些 Agent 神文没告诉你的事：照着做，系统只会更烂 【AI ag

					ent 搭建实操指南】 HL: ⇒ Sure not everything is AI, instead most of ity is not AI Character ≠ TTS voice — it is a full voice recipe
			• AI recommends best fit character (★ Suggested), according to entity Configuration	Phase 3	POC needed
		Voice Selection	• Filter by: All / Female / Male • AI recommends based on entity configuration (non-AI, only by logic) • Available: Aoede, Algieba, Algenib, Despina, Laomedeia, Pulcherrima, Sadaltager, Sulafat	Phase 1	Handle in Studio App (Wizard) and ADK Gemini 2.5 Pro TTS voices

		Context / Director Note	• Optional situational directive • AI generates: Vocal Environment, Mission of Speech, Pacing & Energy • Admin can edit underlying meta-prompt	Phase 1		Handle in Studio App (Wizard) and ADK
		Script Enhancement	• Toggle ON/OFF (default OFF) • AI adds performance cues: non-speech sounds, style modifiers, pacing tags • Max 2 tags per sentence • Client edits enhanced script per language tab and can regenerate	Phase 2		Phase 2 — Epic 8.1
		Audio Generation	• Generate audio per language or all languages • Token estimate shown before generation - Progress bar	Phase 1		Handle in Studio App (Wizard) and ADK Japanese Kanji→Hiraga

			during generation • Japanese: Kanji → Hiragana conversion before TTS • Chinese (TW/CN): pronunciation substitution library		na critical for audio quality
		Audio Playback & QA	• Audio player per language tab: play, pause, skip ±15s • Admin: download .mp3 (24000Hz) • Production: audio written directly to CMS	Phase 1	Handle in Studio App (Wizard) and ADK
		Human Gate 5 — Audio Review	• Client listens per language	Phase 1	Handle in Studio App (Wizard) and ADK
			• Edit enhanced script and regenerate if did script enhancement	Phase 2	
		Audio Pronunciation Fix (<i>POC Needed</i>)	• Client marks a specific timestamp + leaves a text comment	Phase 1	Smart Re-generation in Phase 2

			describing the voice issue Remark that “This is AI function and can make mistakes. Please check carefully before launch.			Feasibility to be confirmed — surgical re-gen requires precise audio stitching
			AI re-generates only that audio segment and blends it back into the full audio - Reference POC built in Google AI Studio: link	Phase 2		HL: Have to try more. So far Google AI Studio version not 100% work
		Generation History	- Every run saved: timestamp, languages, character, context, token count - Click to retrieve scripts and audio for auditing, rollback, and cost tracking	Phase 1		Handle in Studio App (Wizard) and ADK
		SRT Generation	Auto-generate subtitle files	Phase 1		

			synced to audio timing • With the original script as base			
	Guide — Step 5: Publishing	Slideshow Builder (TTML)	• Phase 1 (Manual): Client selects images/videos from Asset Management per spot • Multi-image selection supported per spot - Guide overview: visual timeline editor and adjust image order and sync timing	Phase 1		From This part, v2 now in Decap CMS. so what is the plan to merge it to Laxy Studio? Phase 1 = manual selection
			• AI auto-generates and sequences images synced to audio per spot (AI Layout Generation)	Phase 2		Phase 2 = AI layout generation — Epic 8.4
		CMS Publishing	• Final content check before publish • Bundle approved content (audio, scripts, images, SRT)	Phase 1		Final pre-publish verification step

			• Push to Firebase CDN			
		Preview	• Client previews final Player experience • Select / confirm Player template	Phase 1		
		Final Approval & Publish	• A preview in mobile view shows • Final review all page and content	Phase 1		
	Guide — Quick Process <i>(Phase 2)</i>	Quick Process Mode	• One-click batch using Tenant default settings (Character, Voice, Context) • Skips Human Gates 2, 3, 4 • Auto-generates all language variants - Pushes to "Pending Review" for final Gate 5 only	Phase 2		Phase 2 — Epic 8.5. Defaults configured in Tenant Information
	Publishing & Delivery	QR Code Generation	• Auto-generate QR code on publish • Downloadable as PNG	Phase 1		Short URL critical for clean QR
		Shortlink / URL	• Generate branded short	Phase 1		Critical dependency

			URL (laxy.click) • Custom slug option			for QR quality
		Embed Code	• Generate embed snippet for third-party websites	Phase 3		
		Unpublish / Archive	• Deactivate live experience • Archive with data retained	Phase 2		
	Analytics & Insights	Experience Analytics	• Views, unique visitors, completion rate • Per-spot engagement: time spent, drop-off, audio play rate • Script variant popularity	Phase 1		Phase 2 — POC required before committing to build Take a look on cost
			• GA Integration, POC Needed): Connect with Google Analytics (each Player page has a GA seed) to pull and display GA data inside Studio dashboard • AI-assisted analysis layer: summarise	Phase 2		

			insights and surface recommendations ⚠ Cost investigation needed: analyse data in Laxy Studio vs. leverage Google Analytics native tools and select more cost-efficient path			
		Export Report	Download as CSV or PDF	Phase 1		
	Billing & Subscription	Subscription Plan	- Manual plan setup - View current plan and limits (experiences, storage, users) - Feature availability by tier	Phase 1		handle in FireCMS no idea the scope of phase 1
			Integrate Payment gateway	Phase 3		
		Plan Upgrade / Downgrade	Self-serve plan change with prorated billing	Phase 3		
		Usage Meter	Real-time quota usage display	Phase 2		

		Invoice & Payment History	• View and download invoices	Phase 2		
	Notification	Email Notification	• New user invite, publish success, billing alerts	Phase 2		

Phase Summary

Phase	Module / Feature
Phase 1 (MVP)	Infrastructure, Super Admin, Member Management, Asset Management, Entity Config, Guide Steps 1–5 (core, manual), Slideshow Builder (manual), Publishing & Delivery
Phase 2	Deep Research, Script Enhancement, SRT, Slideshow Builder (AI-Assisted / AI Layout Generation), Quick Process, GA Analytics Integration
Phase 3	Chatbot, Q&A, Stamp Hunt
POC Needed	Audio Pronunciation Fix (segment re-gen + stitch), GA Analytics AI Analysis Layer

Open Items

1. **Player PWA** — Separate function list table.

2. **Voice list** — Finalise available voices per Kazu's research.
3. **Voice Cloning** — Future feature. Licensing required — Alex to confirm with sales team.
4. **Japanese Kanji → Hiragana** — Critical pre-processing step for audio quality. Done transparently server-side.
5. **Chinese Pronunciation Library** — Maintainable substitution table for ZH-TW / ZH-CN edge cases.
6. **Shortlink / URL Service** — Critical for clean QR codes. Third-party service to be selected.
7. **Audio Pronunciation Fix (POC)** — Segment-level audio re-generation and stitching. Feasibility TBD. Reference POC built in Google AI Studio by Harold.
8. **Deep Research Degree UX** — Define exact UI for Low / Medium / High selection. Warning copy to be reviewed by Alex.
9. **GA Analytics Integration (POC)** — Investigate cost: build in-Studio analytics pipeline vs. surface GA data directly. Decision to be made before Phase 2 build.